MILITARY SPECIFICATION

PROPELLANT, LIQUID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a type of liquid propellant containing Ethylene Glycol Dinitrate and 1-Nitro-2-Propanol (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids on request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-N-3399 -	2-Nitrodiphenylamine
MIL-E-48225-	Ethylene Glycol Dinitrate (EGDN)
MIL-N-48263-	1-Nitro-2-Propanol (NPL)
MIL-A-48078-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipping and Storage
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E300 - Sampling Industrial Chemicals
ASTM-D1475 - Density of Paint, Varnish, Lacquer and Related products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Liquid Propellant. The liquid propellant shall conform to the following requirements.

3.1.1 Materials. The constituent materials used in the manufacture of the liquid propellant shall conform to the following specifications:

MIL-N-3399 - 2-Nitrodiphenylamine
MIL-E-48225 - Ethylene Glycol Dinitrate (EGDN)
MIL-N-49263 - 1-Nitro-2-Propanol (NPL)

3.1.2 Composition. The propellant composition on a moisture free basis, shall be as specified Table I.

TABLE I

Constituents	Percent	Test Paragraph
Ethylene glycol Dinitrate	87.1 ± 1.5	4.5.1.1
1-Nitro-2-Propanol	12.0 ± 1.0	4.5.1.2
2-Nitrodiphenylamine	0.9 ± 0.15	4.5.1.3

3.1.3 Moisture. The moisture shall be 0.5 percent maximum, when tested as specified in 4.5.2.

3.1.4 Specific gravity. The specific gravity shall be 1.44 at 22°C minimum, when tested as specified in 4.5.3.

3.1.5 Stability. The stability shall be 8 minutes, min, when tested as specified in 4.5.4.

3.1.6 Heat of Combustion (EC) and heat of explosion (Q). The heat of combustion shall be 2072 ± 40 cal g^{-1} and the heat of explosion shall be 1484 ± 25 cal g^{-1} when tested as specified in 4.5.5.

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3.2 Workmanship. The liquid propellant shall be a clear liquid free from suspended particles when examined as specified in 4.5.6.

3.3 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 1 pt of liquid propellant obtained by sampling as described in 4.4.2.1. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same source of supply as will be used in regular production.

4.3.2 Inspections to be Performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. For the material covered by this specification, a lot shall consist of one or more batches of liquid propellant produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of liquid propellant that has been subjected to the same unit chemical or physical process.

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4.4.2 Testing. Caution is to be exercised in handling the sample due to its explosive sensitivity and toxicity (see 6.3).

4.4.2.1 Sampling. Sampling shall be accomplished from the mixing tank. The propellant shall be sampled in accordance with paragraph 7 of ASTM-E300 using a rubber dipper. Two (2) independent representative samples of approximately 2 ounces shall be removed from the vessel. Transfer the sample to a rubber bottle with a rubber stopper. The bottles shall be labelled to show the following:

a. sample number
b. date manufactured
c. batch identification number

If either sample fails to comply with any of the requirements specified, the batch shall be rejected. The samples shall be tested for the following properties:

Inspection/Requirement	Defects Classification
Composition (see 3.1.2)	Major B
Moisture (see 3.1.3)	Major B
Specific Gravity (see 3.1.4)	Major B
Stability (see 3.1.5)	Major B
Heat of Combustion and Heat of Explosion (see 3.1.6)	Major B

4.4.2.2 Decontamination of samples bottles. The bottle and stopper shall be cleaned as follows (see 6.3).

(a) Rinse thoroughly with acetone
(b) Flush generously with hot tap water (3 minutes under a direct flow is considered adequate)
(c) Rinse with distilled water and dry

4.4.3 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4 and 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpose.

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4.5 Test Methods and Procedures. The tests in 4.5.1 thru 4.5.6 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks. (see 6.2)

4.5.1 Composition

4.5.1.1 Ethylene glycol dinitrate. The ethylene glycol dinitrate shall be determined in accordance with Method 208.1 of Standard MIL-STD-286 using the factor 2.534 in the calculation and the weight of sample corrected for a moisture free basis.

4.5.1.2 1-Nitro-2-Propanol. The 1 Nitro-2-propanol shall be determined by difference.

4.5.1.3 2-Nitrodiphenylamine. The 2-Nitrodiphenylamine shall be determined in accordance with Method 218.4 of Standard MIL-STD-286 and the weight of sample used corrected for a moisture free basis.

4.5.2 Moisture. The moisture shall be determined in accordance with Method 101.5 of Standard MIL-STD-286.

4.5.3 Specific gravity. The specific gravity shall be determined in accordance with ASTM-D1475.

4.5.4 Determination of stability. The stability shall be determined as follows: Filter a portion of the propellant sample through two thicknesses "S and S" Number 604 filter paper or equivalent. Transfer a 2 ml portion of the filtered material by means of a pipette, to each of three test tubes which are 5.5 inches long, 0.50 inches internal diameter and 0.62 inches outside diameter. (Care should be taken during the transfer not to leave droplets of the propellant on the sides of the test tubes). Stopper each tube by means of a new, tightly fitting cork through which passes a tightly filled glass rod equipped with a platinum holder for a strip of standard potassium iodide starch indicator paper. Using a forcept hang on the platinum holder a strip of standard potassium iodide starch indicator paper. The standard test papers shall be furnished by the Naval Propellant Plant through the contracting activity and shall normally be approximately 1 inch long and 3/8 inch wide. Moisten a horizontal section of the upper half of the standard test paper with a 50 percent solution (by volume) of pure glycerin in distilled water. This is conveniently accomplished by dipping a small diameter glass rod into the glycerin water solution, and as the rod is withdrawn, contact is made with the side of the container so as to minimize the volume of the solution adhering to the rod.

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Draw the rod across the paper strip so as to produce a level and distinct line of demarcation on the lower edge of the wet area. Prepare a blank by suspending a moistened strip of the standard test paper in a clean dry tube. Adjust the temperature of the heat tube batch to 82.2 + 1°C. and insert all four test tubes into the heating solution. The depth of immersion of the test tube into the heating solution should be approximately 2 inches. The bath shall be placed in such a position that the test tubes are viewed against a white background illuminated by bright diffused daylight. Note the time of insertion of the tubes into the bath. During the test the line of demarcation on the standard test paper should be 3 inches above the level of the propellant in the test tube. The line of demarcation in the blank tube is regulated at an equivalent height in the tube. Consider the end point of the test to be first appearance of any discoloration at the line of demarcation between the wet and dry portion of the test paper in the sample tube which is in excess of the discoloration observed at the same position on the test paper in the blank. Note the time for completion of the test to the nearest minute. Consider the minimum time for the three tubes to represent the heat value of the sample. After the test rinse all rods and tubes with acetone to remove propellant wash with warm soapy water, rinse thoroughly with tap water, then with distilled water and dry in a steam oven at approximately 80°C.

4.5.5 Heat of Combustion and Heat of Explosion. A standard adiabatic combustion calorimeter shall be calibrated with benzoic acid, calorific value 6318 cal/g^{-1}. The heat of combustion and the heat of explosion shall be determined in 25 atmospheres of oxygen and helium respectively. One gram sample shall be used in each test and a series of 5 or more determinations shall be made on each sample. All results shall fall within the range, see 3.1.6. The sample shall be ignited with six inches of 28 B and C gauge nichrome wire, which shall be wound in a coil (10 loops) and placed in the liquid. The ignition energy shall be measured with a watt second meter. An iron fuse wire shall be used for the heat of explosion test. Detail description of the calorimeter test shall be determined in accordance with the "Parr" Company Manuals and procedures described by F.D. Roosini "Experimental Thermo-Chemistry" Inter Science Publishers Inc. N.Y. 1956.

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4.5.6 Workmanship. Pour a 100 ml sample into a 100 ml cylinder and visually examine the liquid propellant for the presence of particles or turbidity.

5. PREPARATION FOR DELIVERY

5.1 Packing. The liquid propellant described in this specification is only intended for intraplant shipment. Movement within the loading plant shall be in accordance with fully approved safety procedures.

6. NOTES

6.1 Ordering Data - See MIL-A-48078 (PA).

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801. This description should include but not be limited to the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Toxicity. The EGDN in the propellant may cause dilation of blood vessels, headache, nausea, cyanosis, and reduce blood pressure. Extreme exposures resulting from inhalation, skin contact or ingestion may result in central nervous system depression, coma, and respiratory paralysis. Drinking of any quanity of the liquids must be avoided. Severe headache shortly after exposure may diminish after development of tolerance. Ventilation should be used to reduce the inhalation hazard. Protective gloves and aprons are helpful in avoiding skin contact. Frequent washing in soap and water will minimize skin absorption. Alcohol or other solvents should not be used on the skin. The Threshold limit value ceiling is 0.2 ppm.

6.4 Intended use. The material described in this specification is intended for use as a propellant in the ADAM.

Custodian:
Army - PA

Preparing Activity:
Army - PA

Project Number: 1376-A021

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